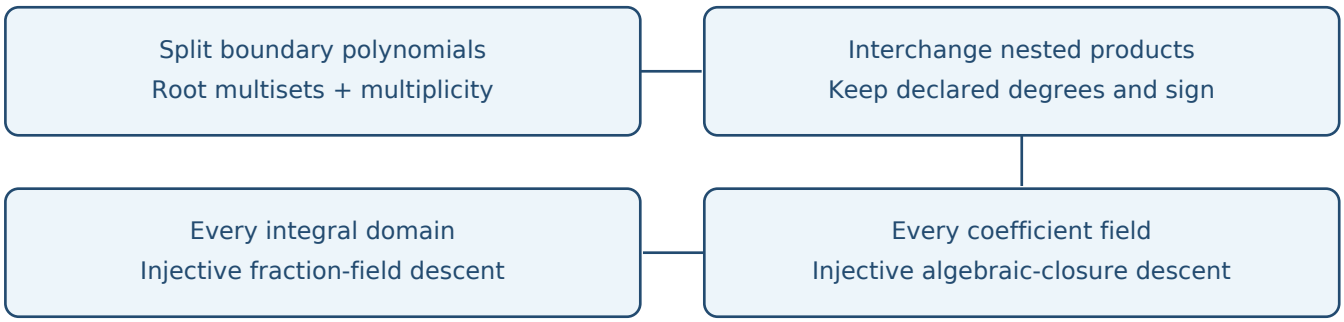
Resultant interchange survives degree drops

Proved component. Over every commutative integral domain, iterated fixed-size Sylvester resultants commute under explicit bidegree bounds. Both intermediate degree bounds are derived. Signed standing: unverified. [pcc:unverified]

Write $a=\deg P(y)$, $c=\deg R(x)$, and $\deg_x Q \le dx$, $\deg_y Q \le dy$. Sizes after the semicolon are declared sizes, retained after specialization. [pcc:unverified]

$\mathrm{Res}_x(\mathrm{Res}_y(P,Q;\, a, dy), R;\, a\cdot dx, c)$
$= \mathrm{Res}_y(P, \mathrm{Res}_x(Q,R;\, dx, c);\, a, dy\cdot c)$. [pcc:unverified]

No nonzero, monicity, splitting, separability, characteristic or intermediate degree-preservation assumption is added. Natural outer sizes also work if both intermediate degree bounds are sharp; that condition is explicit in a separate theorem. [pcc:unverified]



Direct negative control over $\mathbb{Q}$. $P=y^2-y$, $Q=xy+1$, $R=2x+1$. Both Q bidegrees are1. The inner polynomials are $x+1$ and $y-2$; the first has degree1 below formal bound2. [pcc:unverified]

Outer sizes	Left	Right
Natural degrees	-1	2
Retained bounds	2	2

Scope. This refutes an unrestricted helper that resets degrees. It does not refute Semaev's qualified theorem or establish a failure of its canonical family. Actual arbitrary-split normalization, higher symmetry and higher absolute irreducibility remain separate obligations. [pcc:unverified]

Evidence. Whole Lean gate: 46 declarations, 0 failures, 0 trusted-native. Lean4.35.0 / Mathlib56a00585…. Standard foundations only; no holes, custom axioms, native evaluation, raised limits or waivers. See sources.json, proof-map.json and evidence/ for exact source/attempt/gate pins. [pcc:unverified]

Sources. Semaev2004 Theorem1, p2 arbitrary-split formula; Mathlib Resultant/Basic and Polynomial/Bivariate/BigOperators; repository DeclaredResultantRoots, ResultantDegreeStep and RecursiveSemaevDegree/Symmetry. KB unavailable; bounded fallback, no novelty claim. No native experiment, paid compute or agent upload. [pcc:unverified]